DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY · SCHOOL OF QUANTUM TECHNOLOGY

# BB84 Protocol Audit Report

Quantum Key Distribution — Full Session Transcript & Security Analysis

QKD-8824-A

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SECURITY VERDICT

**SECURE — KEY ESTABLISHED**

EAVESDROPPER (EVE)

✅ NOT DETECTED — CHANNEL SECURE

## §1 PROTOCOL CONFIGURATION

BB84 QKD — Simulation Parameters

| | |
|---|---|
| **Protocol** | BB84 Quantum Key Distribution (Bennett & Brassard, 1984) |
| **Photon Bases** | Rectilinear basis (+): $|0\rangle$, $|1\rangle$ \| Diagonal basis (×): $|+\rangle$, $|-\rangle$ |
| **Photons Transmitted (n)** | 1000 |
| **Channel Noise Level** | 6.0% (Noisy Channel) |
| **Eavesdropper Active** | No — Channel not compromised by eavesdropper |
| **Error Correction Protocol** | LDPC (Belief Propagation) Protocol |
| **Privacy Amplification** | Universal Hashing via Toeplitz Matrix Multiplication |

### INFORMATION-THEORETIC SECRECY AUDIT

Based on the detected **QBER (4.65%)**, the simulator calculated a partial information leakage to Eve of approximately **31.2%**. Privacy Amplification was applied using a **Toeplitz Hash Function** to map the 247-bit working key into a compressed space of **169 bits**.

ENTROPY RETENTION PROFILE

68.4%

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## STAGE 3 — QUANTUM TRANSMISSION

Alice prepares photons → quantum channel → Bob measures

| PHOTONS PREPARED | ALICE BITS | ALICE BASES | BOB MEASUREMENTS |
|---|---|---|---|
| 1000 | 1000 | 1000 | 1000 |
| Alice → Quantum Channel | Raw random bit sequence | + and × encodings | Received qubit readings |

**ALICE RAW BITSTREAM** — 1000 BITS

```
11000101 00000111 00011000 10100100 10000000 10010010 01010100 01010100 10001011 0
0011101 00111100 10100110 11100100 01101010 11001101 10111111 … +872 more bits
```

**ALICE BASES (+/×)** — 1000 BASES

```
×+×++×++ ++×+×××× +++×+++× ++×××+×× +×+×++++ +×××+×++ ×××+×××× ×+×+++×× +++++×+× ×
++×××++ ++++××+× +××+×+×× +××××+++ ++×++×++ ×+++++×+ ++×+×+++ … +872 more
```

**BOB BASES (+/×)** — 1000 BASES

```
××+××××+ ++××+++× +×+××+×+ ++×+×+++ +×+++×+× ×++×++×× ××+×+××+ +++×++×+ +××+++++ ×
+××++×+ ××+××××× ××××+++× ×+××++++ +×+++×+× ×++×++++ +××+××+× … +872 more
```

**BOB MEASURED BITS** — 1000 BITS

```
10011111 00000001 00011000 10110100 10011001 11100001 01111100 01110101 11101010 0
0111001 10101110 00110110 11101100 01101011 11100111 10111010… +872 more bits
… +872 more bits
```

**Quantum Channel Integrity:** No eavesdropper was inserted. Alice transmitted 1000 photons over the quantum channel. Bob received and measured each photon independently using a randomly chosen basis (with 6.0% channel noise applied).

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## STAGE 4 — BASIS SIFTING

Alice and Bob publicly compare bases; keep only matching-basis bits

| RAW BITS (INPUT) | SIFTED KEY LENGTH | BITS DISCARDED | SIFTING RATE |
|---|---|---|---|
| 1000 | 519 | 481 | 51.9% |
| Pre-sifting | Basis-matched bits | Basis mismatch | ~50% expected |

**ALICE SIFTED KEY (BITS AT MATCHING-BASIS POSITIONS)** 519 BITS

```
11100010010101011000001101011010110011111001010101000101011101111111101010110111010
1111111001100111100110111111000000010000001111101 …+391 more
```

**BOB SIFTED KEY (SHOULD MATCH ALICE — ERRORS APPEAR BELOW)** 519 BITS

```
11100010010101011000001111011010110011111001010101000101011101111111101010111111010
0101111001100111000110111111000000010010011111101 …+391 more
```

**Protocol Note:** Sifting retains ~50% of raw bits (both parties chose the same basis). In this session, 481 bits were discarded due to basis mismatch, yielding a sifted key of 519 bits. Eve's interception would not be visible at this stage — errors only appear after QBER estimation.

## §4 STAGE 5 — QBER ESTIMATION

Quantum Bit Error Rate — Channel Security Parameter Estimation

| SIFTED POOL | BITS SACRIFICED | QBER | WORKING KEY |
|---|---|---|---|
| 519 | 129 | 4.65% | 247 |
| Input bits | For QBER sample | Threshold: 12.9% | Post-sacrifice bits |

QBER LEVEL

✅ NOT DETECTED — CHANNEL SECURE

0% | 12.9% Limit | 100%

| PARAMETER | VALUE | INTERPRETATION |
|---|---|---|
| **QBER (Q)** | 4.65% | Non-zero but within tolerance — channel noisy |
| **Security Threshold ($Q_{max}$)** | 12.9% | BB84 standard: abort if Q > 12.9% (Hoeffding Bound) |
| **Bits Sacrificed for QBER** | 129 | ~25% of sifted key publicly compared for error detection |
| **Remaining Working Key** | 247 | Proceeds to error correction phase |
| **Eve Detection Confidence** | **LOW — Noise present, possible natural errors** | Based on Hoeffding Inequality (statistical bound) |

● ALICE WORKING KEY — POST QBER SACRIFICE

247 BITS REMAINING

```
1010111001111100110101010101101111100000000010000010010001100111101101100100001010
10001011111001010000000111010111011111111010011 …+119 more
```

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## STAGE 6 — POST-PROCESSING

Error Correction (LDPC (Belief Propagation)) + Privacy Amplification (Toeplitz Hashing)

| INPUT BITS ($R_{IN}$) | LEAKAGE ($I_E$) | PA FACTOR | NET SECRECY |
|---|---|---|---|
| 247 | 31.2% | 0.68x | 169 |
| Post-Reconciliation key | Eve's potential info | Compression ratio | Final secure bits (m) |

| STAGE | PROTOCOL | BITS IN | BITS REMOVED | BITS OUT |
|---|---|---|---|---|
| **Raw Transmission** | BB84 Photon Encoding | – | – | 1000 |
| **Basis Sifting** | Public basis reconciliation | 1000 | 481 | 519 |
| **QBER Sampling** | Statistical error estimation | 519 | 129 | 247 |
| **Error Correction** | LDPC (Belief Propagation) | 247 | 143 | 104 |
| **PA Leakage Purge** | Entropy Nullification ($ncdotH(Q)$) | 104 | 67 | 37 |
| **PA Safety Margin** | Finite-Key Safety Hash | 37 | 11 | 169 |

**SECRECY CAPACITY BOUND**

$$l \leq n - \text{leakage}_{EC} - H_{max}(X|E) \implies l \approx n \cdot (1 - f \cdot H_2(Q))$$

The final key length is mathematically constrained to ensure that even with infinite computing power, the eavesdropper (Eve) possesses $le2^{-k}$ bits of information about the secret key.

**Error Correction (LDPC (Belief Propagation)):** Alice and Bob reconcile their keys by exchanging parity information over a public classical channel — revealing only whether blocks of bits have even or odd parity, not the bits themselves. This eliminates the 143 bits introduced by channel noise.

**Privacy Amplification (Toeplitz Hashing):** The key is compressed using a randomly chosen Toeplitz matrix. Any partial knowledge Eve acquired during Error Correction (from public parity broadcasts) is provably eliminated by this compression, resulting in an information-theoretically secure final key.

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## STAGE 7 — FINAL SECRET KEY

Unconditionally secure shared secret — ready for ChaCha20-Poly1305 encryption

FINAL SECRET KEY — ESTABLISHED QUANTUM-SECURE SYMMETRIC KEY

169 BITS

1000100010011111011111010010011001101111011010111111000101111100
001101011001101111011111101110100001011011111001100010111000010
00 …+41 more bits

DIGITAL KEY FINGERPRINT (SHA-HASH SIMULATION)

889F-7D26-6F6B-F178-6B37-BF74-2DF3-1708-70CC

ENTROPY RETENTION

**16.9%**

| FINAL KEY LENGTH | KEY GENERATION EFFICIENCY | PROTOCOL STATUS |
|---|---|---|
| **169** | **16.9%** | 🔐 **SECURE** |
| Secure bits | Final / Raw ratio | All security proofs satisfied |

| RAW | | SIFTED | | WORKING | | FINAL |
|---|---|---|---|---|---|---|
| **1000** | → | **519** | → | **247** | → | **169** |
| Photons | | Basis Match | | Post-QBER | | Secret Key |

🔐 **Key Ready:** Alice and Bob now share an identical, unconditionally secure 169-bit secret key. In practice, this key would be split: half for **ChaCha20** (stream cipher — ensures confidentiality) and half for **Poly1305** (MAC — ensures message authentication and integrity).

## §7 SECURITY ANALYSIS SUMMARY

Information-theoretic security assessment of this QKD session

| SECURITY PARAMETER | VALUE / STATUS | REQUIREMENT | RESULT |
|---|---|---|---|
| **QBER** | 4.65% | Q ≤ 12.9% | **Pass** |
| **Eve Detection** | ✅ NOT DETECTED — Channel Secure | No eavesdropper | **Pass** |
| **Key Established** | 169 bits | > 0 bits | **Pass** |
| **Key Efficiency** | 16.9% | Protocol dependent | **Normal** |
| **Error Correction** | LDPC (Belief Propagation) | Cascade / LDPC | ✅ **Applied** |
| **Privacy Amplification** | Purge: 67 bits | $H(Q)$ Correction | ✅ **Leakage Nullified** |
| **Key Fingerprint** | 889F-7D26-6F6B... | Digital Signature | **Verified** |
| **Overall Verdict** | 🔐 **SECURE — KEY ESTABLISHED** | | |

**Academic Context:** This report documents a **classical simulation** of the BB84 protocol. All quantum state measurements are performed probabilistically using pseudo-random number generation. Real-world QKD requires photonic hardware including single-photon sources, polarising beam-splitters, and single-photon detectors (SNSPDs). This simulator is designed for *pedagogical demonstration* of quantum cryptographic principles at DIAT's School of Quantum Technology.

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